



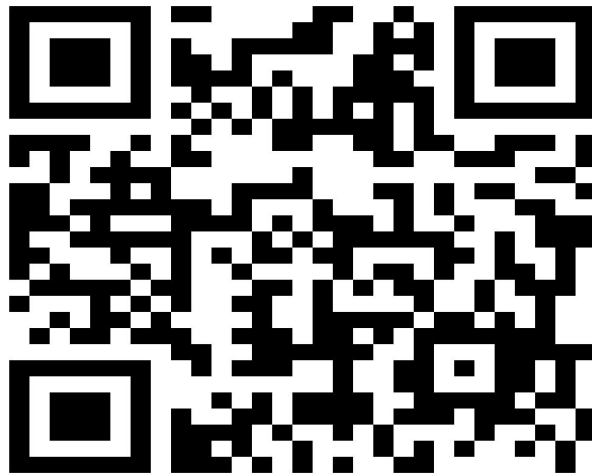
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TEST TIME ON DATA SUFFICIENCY

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CASELET PROBLEMS



DATA CASELET

What is a Data Caselet?

A Caselet is a set of information that is given in paragraph form. In a Caselet, no tables or graphs are given. You have to read the information and draw a table or organize the given data to solve the questions. Practice a lot of caselets based questions to master such questions.

5 Tips for Data Caselets

- ❖ Use these five simple tips to ace data caselets: It helps to convert a caselet into table or any other pictorial form.
- ❖ Keep jotting down the important points in a caselet while reading it for the first time. Use notations to represent names of things, places, people, etc.
- ❖ Be careful of the data given and the units as well. You need to be aware of basic conversions and units. For example, what do the terms million/billion/trillion stand for.

5 Tips for Data Caselets

- ❖ Make sure you use the options and approximations to avoid lengthy calculations.
- ❖ Cover arithmetic related topics (interest, percentage, ratio & proportion) as questions in caselets are often based on concepts from this area.
- ❖ Let's solve a solve caselet now and see how to approach these questions.



Directions for 1-5

Mlon husk , the space Y inventor took his team of scientist to newly found planet "Euro Chand". The team agreed on the condition that they will not do any research during their visit but will only do rest. But, Husk as per his nature observed various phenomenon's on new planet, including the gravitational pull. He observed that a kg. on earth is equivalent to 12 kg on Euro Chand , except kg on new planet is called Euroc. Slen weighs 552 Euronc,weighs heavier than all except Plen,who weighs 108 Euronc more. Olen, the lightest, weighs 252 Euronc less than Dlen. Jlen weighs 48 Euronc less than Slen and 48 Euronc more than Dlen.

Question: 01

What is the weight of Plen (in Kg)?

- A. 55
- B. 46
- C. 17
- D. 48

Answer: A

Explanation: 01

Solution:

Kg Euronc

Jlen 42 504

Dlen 38 456

Plen 55 660

Slen 46 552

Olen 17 204

As shown above the weight of Plen is 55Kg.

Question: 02

What is the weight of Olen (in Euronc)?

- A. 505
- B. 246
- C. 218
- D. 204

Answer: D



Explanation: 02

Solution:

	Kg	Euronc
Jlen	42	504
Dlen	38	456
Plen	55	660
Slen	46	552
Olen	17	204

As shown above the weight of Plen is 55Kg.

Question: 03

Approximately how many times is the weight of Plen that of Dlen?

- A. 1.45
- B. 1.07
- C. 2.18
- D. 2.14

Answer: A

Explanation: 03

Weight of Plen is $55/38=1.45$ that of Dlen

Question: 04

Weight of Dlen is what % more/less than weight of Jlen?

- A. 9.5
- B. 10.8
- C. 11.1
- D. 12.1

Answer: A



Explanation: 04

Required percentage = $42 - 38 / 42 \times 100 = 9.5\%$

Question: 05

What is the weight of Jlen(kg)?

- A. 42
- B. 35
- C. 52
- D. 40

Answer: A

Explanation: 05

Solution:

	Kg	Euronc
Jlen	42	504
Dlen	38	456
Plen	55	660
Slen	46	552
Olen	17	204

As shown above the weight of Plen is 55Kg.



Directions for 6-10

Refer to the data below & answer the questions that follow.

A man goes on dividing a certain sum of money into half. He does this 9 times. He gives the amount in round 1 and round 4 and round 7 to A. To B he gives the amount in rounds 2, 5 and 8. To C he gives the amount in Rounds 3, 6 and 9. Assume that the balance sum at the end of Round 9 is Rs. 195.31.

Question: 06

What is the original sum of money?

- A. Rs. 10×10^4
- B. Rs. 10^4
- C. Rs. 10^6
- D. Rs. 2.5×10^5

Answer: A



Explanation: 06

If the original amount is x at the end of round 9, the amount left is $x / 2^9 = x / 512$

Since the amount is Rs. 195.31 the original sum, $x = 195.31 \times 512$

Approximately $200 \times 500 = 10 \times 10^4$.

Question: 07

What is the ratio of shares of A : B : C ?

- A. 10 : 5 : 2
- B. 12 : 6 : 2
- C. 4 : 2 : 1
- D. 1 : 1 : 1

Answer: C

Explanation: 07

Ratio of shares cannot be equal.

The ratio has to be such that share of C is half of B which is half of A. So answer is 3.

Question: 08

If A gets amounts in Rounds 1, 2 and 3, ; B in rounds 4, 5 and 6 ; C in Rounds 7 , 8 and 9
what is the ratio of shares of A : B : C ?

- A. 64 : 8 : 1
- B. 256 : 128 : 4
- C. 128 : 64 : 8
- D. 12 : 4 : 1

Answer: A



Explanation: 08

A:B:C =

$$(1/2 + 1/4 + 1/8) : (1/16 + 1/32 + 1/64) : (1/128 + 1/256 + 1/512) \\ = 64:8:1$$

Question: 09

If the balance at the end of Round 9 is Rs. 390.625, the original sum is

- A. Rs. 2.5×10^4
- B. Rs. 2×10^5
- C. Rs. 2×10^6
- D. Rs. 3×10^8

Answer: B

Explanation: 09

If the sum at the end of round 9 is Rs. 195.31, original sum is 10. If amount is Rs. 390.625, then the original amount is 2×10^5 .

Question: 10

How much does A get in all (nearest 100)?

- A. Rs. 45100
- B. Rs. 48100
- C. Rs. 57000
- D. Rs. 62000

Answer: C



Explanation: 10

A gets $x/2 + x/16 + x/128 = (64 + 8 + 1)x/128$
 $= 73x/128$ is approximately 57,000.
Hence Option (3).



Directions for 11-13

Nizam tailor, the famous tailor of Naharpur, after facing losses for 3 consecutive years diversified his business. He started taking orders for school uniform. Initially he got orders from two schools: Moti Ram School and Lala Hardy School.

Nizam hired two cutters, five tailors and two helpers for this. Cutters cut the cloth, tailors stitch the clothes and helpers do the finishing work like buttoning etc .All employees work for ten hours a day. Timings for making uniform for both schools are different. The details are given below:

	Cutting time	Stitching time	Finishing time (buttoning etc)
MOTI RAM SCHOOL	20 Min	1Hour	15 Min
LALA HARDYAL SCHOOL	30 Min	1 Hour	30Min



Question: 11

One day Nizam decided to make 25 uniforms of Moti Ram School only, the man hours that will remain idle is(approx.)

- A. 30
- B. 50
- C. 55
- D. 26

Answer: B



Explanation: 11

Time taken to make 25 uniforms of MOTI RAM SCHOOL Time taken by cutter = $25 \times 20 = 500$ min. Time taken by tailors = $25 \times 60 = 1500$ min. Time taken by helper = $25 \times 15 = 375$ min. So man hours that are idle = $(1200 - 500) + (3000 - 1500) + (1200 - 375) = 700 + 1500 + 825 = 3025$ = $3025/60 = 50$ hours (approx)

Question: 12

If he hires two more cutters and one more tailor, then what is the maximum number of Moti Ram School uniform can be made by him in a day?

- A. 120
- B. 70
- C. 80
- D. 60

Answer: D

Explanation: 12

For MOTI RAM SCHOOL

Total cutters now = $2+2=4$

Number of uniforms cut by cutters = $2400/20 = 120$

Total tailors now = $5+1=6$

Max Number of uniforms stitched by tailors = $3600/60 = 60$

Max Number of uniforms finished by helpers = $1200/15 = 80$

Hence limiting factor is uniforms stitched by tailors. So 60 is the answer.

Question: 13

If he hires two more cutters and two more tailors, then what is the maximum number of Lala Hardy school uniform can be made by him in a day?

- A. 120
- B. 40
- C. 80
- D. 60

Answer: B

Explanation: 13

For LALA HARDYAL SCHOOL

Total cutters now = $2+2 = 4$

Number of uniforms cut by cutters = $2400/30 = 80$

Total tailors now = $5+2 = 7$

Max Number of uniforms stitched by tailors = $4200/60 = 70$

Max Number of uniforms finished by helpers = $1200/30 = 40$

Hence limiting factor is for helpers. So 40 is the answer.

Directions 14-15

The "Product Tree" of a consumer good X is as follows:

1. Unit of X is made up of two units of A, three units of B and two units of C. 2. Each unit of A is made up of raw material M of 3 units. 3. Each unit of B is made up of 1 unit of P and Q each. One unit of P is made of 2 units of M. One unit of Q is made up of 2 units of A. 4. One unit of C is made up of three units of D, each of which consumes two units of M.

Question: 14

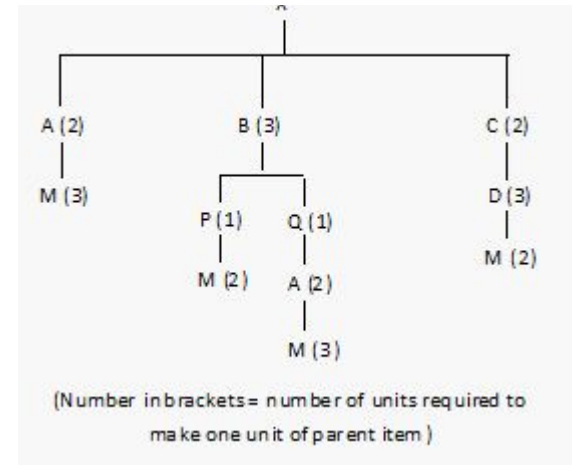
If market demand for product X is 500 units and each of the unit of M costs Rs. 2.5, what is the total raw material cost?

- A. Rs. 58000
- B. Rs. 62500
- C. Rs. 52500
- D. Rs. 35000

Answer: C

Explanation: 14

- i) amount of M required for A = $2 \times 3 = 6$
- ii) amount of M required for B
 $= [2 \times 1 + 3 \times 2 \times 1] \times 3 = 24$
- iii) amount of M required for C = $2 \times 3 \times 2 = 12$
 $\therefore$ One unit of X consumes 42 units of M.
 $\therefore$ Total raw material cost = $42 \times 500 \times 2.5$
 $= \text{Rs. } 52,500.$
Hence, [3].



Question: 15

The raw material cost constitutes of 70% of the total variable cost and the manufacturer incurs a fixed cost of Rs. 1 lac for producing X. If he decides to sell X at Rs. 250 per unit, how many units of X should he produce to break even his costs?

- A. 1000
- B. 1500
- C. 2000
- D. 2500

Answer: A



Explanation: 15

IMAGE-><https://mba.hitbullseye.com/sites/default/files/mobilecommonImages/Data-Caselets-Practice-Questions-answer.jpg>

From answer to previous question, raw material cost

per unit of X = $52500 / 500$

= Rs. 105

Total variable cost = $105 / 0.7$

= Rs. 150

Selling price = Rs. 250.

To break even, he should cover the total variable cost as well as the fixed cost.

Let n be the number of units produced at break-even.

Then $250n = 1,00,000 + 150n$

$\therefore n = 1,00,000 / 100 = 1000$.

Hence, option 1.





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THANK YOU